

Additive Manufacturing.



**KALINGA INSTITUTE
OF INDUSTRIAL TECHNOLOGY**

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Syed Md Noor Afjal 2202046

**Autumn 2023
3rd Semester**

Acknowledgement

We would like to sincerely thank the distinguished instructors and the dean of KIIT's School of Mechanical Engineering (SME) for their great support and direction. For the entire time that our report on "ADDITIVE MANUFACTURING." Their constant support, knowledge, and guidance have been crucial in forming our comprehension of engineering concepts and encouraging a mindset of inventiveness. We are truly grateful to Dr. SWAYAM BIKASH MISHRA for his perceptive guidance and helpful comments and academic guidance, which greatly aided in the course's successful completion. Their commitment to developing our technical proficiency and creating a cooperative learning atmosphere has been incredibly motivating. Additionally, we are incredibly grateful for the inspiring guidance and assistance supplied by SME's Dean, Sir. Their dedication to cultivating A tradition of experiential learning and academic excellence has been essential in enabling us to take on this difficult attempt and acquire useful knowledge in the field of mechanical technical. We express our gratitude to the SME community as a whole. for giving us access to cutting-edge resources and facilities, and a favourable atmosphere for learning, which have been crucial to the accomplishment of our course objectives. We have learned a great deal from this project, and we are incredibly appreciative of the faculty and Dean of SME for their unwavering encouragement, confidence in our abilities, and support.

Introduction

3D printing, an innovative process for producing objects, creates intricate designs by building up layers from digital models. Additive Manufacturing is a type of 3D printing, also known as rapid prototyping. In contrast to conventional methods, which require removing material, 3D printing utilizes precise additive processes to produce designs of unparalleled complexity and intricacy. This advanced technology holds significant potential for fostering innovation in diverse sectors, including aerospace, healthcare, and automotive, as it enables the creation of intricate, customized structures with unprecedented flexibility.

Application

Additive manufacturing is used in a variety of industries, such as aerospace, automotive, consumer goods, medical, and architecture. In aerospace, additive manufacturing is used to create lightweight and complex parts, reducing material waste and improving fuel efficiency. In automotive, additive manufacturing enables rapid prototyping and customization of parts and lightweight structures for better vehicle performance.

In consumer goods, additive manufacturing helps to create personalized products such as fashion accessories, home décor, and more. In dental, additive manufacturing allows for the fabrication of accurate dental prosthetic and models for surgeries. In tooling, additive manufacturing enables the rapid and cost-efficient production of components for manufacturing processes, as well as the creation of complex and innovative architectural models and structures. Additive manufacturing is used in a variety of industries, including education, electronics, and military and defense. In education, additive manufacturing provides hands-on experience in prototyping and model building. In electronics, additive manufacturing creates complex circuitry and customized electronic components. In military and defense, additive manufacturing is used to quickly prototype specialized equipment and components. These applications illustrate the versatility and transformative impact of additive manufacturing.

Advantages

Additive manufacturing has several key benefits, such as:

Design flexibility: Additive manufacturing enables complex and intricate designs that would be difficult or impossible to create with traditional manufacturing methods.

Personalization: Additive manufacturing allows for the creation of personalized and tailored products, such as medical implants or consumer goods.

Reduced material waste: Additive manufacturing reduces material waste because it adds material layer by layer, unlike subtractive methods.

Rapid prototyping: Additive manufacturing accelerates product development cycles by producing prototypes quickly, reducing time to market.

Cost-effective for low volume production: Additive manufacturing is well suited for small production runs, as it eliminates the need for costly molds or tooling.

Weight reduction: Additive manufacturing creates lightweight structures, which are essential in industries such as aerospace and automotive, for improved efficiency.

Complex Geometries: Able to create internal structures and complex geometries that are hard or impossible to create using conventional approaches. Supply Chain Efficiency: Facilitates Production on Demand, lowering the requirement for large storage facilities and extended supply chains. Improved Product Performance: Makes designs more suitable for certain activities, resulting in enhanced performance across several applications. Material Innovation: Continuous advancements in the field of materials increase the number of uses, such as biocompatible materials with high performance and those intended for medicinal application aircraft. Due of these benefits, additive manufacturing has revolutionary potential. Technology across a variety of sectors, encouraging creativity and effectiveness.

Future Prospects in the Field of Additive Manufacturing.

- a. Material Advancements: Ongoing research aims to develop new materials with enhanced properties suitable for a broader range of applications.
- b. Industry 4.0 Integration: AM is becoming an integral part of the fourth industrial revolution, with smart factories incorporating AM into their production processes.
- c. Medical Breakthroughs: Continued advancements in medical AM applications, including bioprinting, hold promise for the healthcare industry

Solidworks

One of the most potent computer-aided design (CAD) programmes used in engineering and product design is SolidWorks. It is an easy-to-use platform designed by Dassault Systèmes for producing 3D models, simulations, and intricate drawings. SolidWorks is widely known for its parametric design features, which enable users to effortlessly alter designs by adjusting parameters. SolidWorks, which offers an extensive range of modelling, assembly, and simulation capabilities, is an essential tool for product development, enabling accurate and productive design from conception to production.

Features of Solidworks

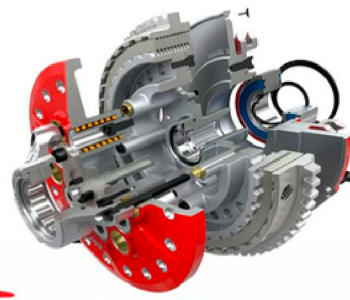
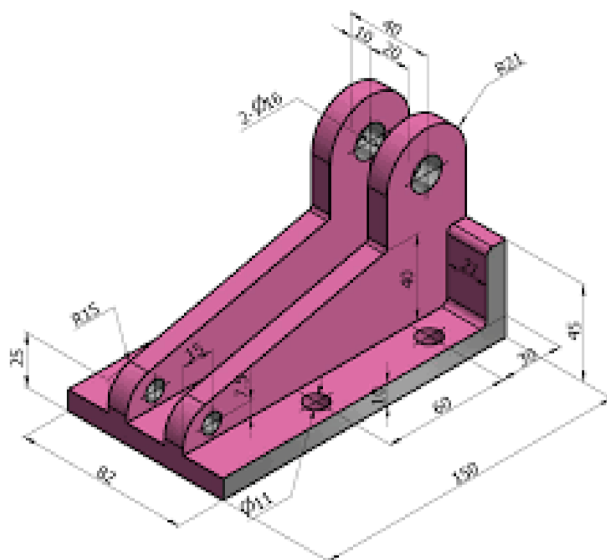
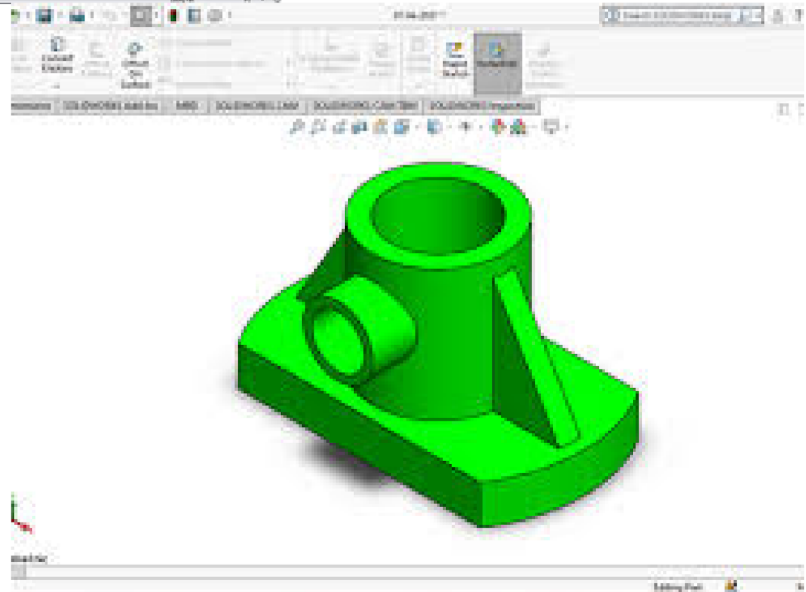
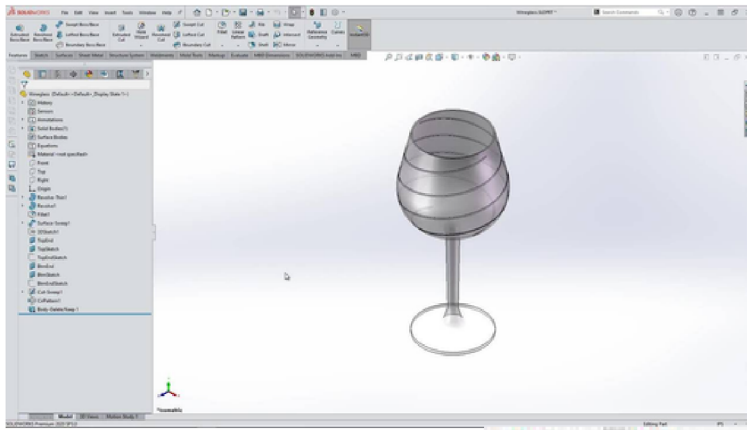
Introduction to SolidWorks: Grasping the essentials of the interface, tools, and fundamental commands.

Sketching and Constraints: Mastering 2D sketching techniques and applying constraints to achieve precision in designs.

Part Modeling: Crafting 3D models of various components with an emphasis on parametric design principles.

Assembly Design: Learning to integrate individual components into intricate structures, with a focus on mates and relationships.

Drawing and Documentation: Generating detailed engineering drawings inclusive of annotations and dimensions.



SOLIDWORKS

Applications of Solidworks

Boost Productivity

SOLIDWORKS 3D modeling enables you to immediately simulate the design aspect in 3D and make any necessary adjustments. Because the designer will not have to rethink the concept later, his efficiency will increase. SOLIDWORKS drawings also aid in the detection of inconsistencies and allow the designer to modify models and sketches before they are completed. This reduces manufacturing time and improves the process, making it more cost-effective and allowing businesses to meet tight deadlines.

Improve Your Design

Traditional drafting methods entail individually recording different aspects of a structural element, which is a time-consuming and accurate process. SOLIDWORKS 3D CAD models come preloaded with a variety of documentation options, such as documenting product configurations and measurements, material specifications, bill of materials, and so on, which simplifies the entire component design documentation process.

Innovation:

By bringing an integrated 3D product development framework such as SOLIDWORKS, advanced manufacturers will achieve the product development quickness, design versatility, and cooperative independence needed to motivate the unique products of the future. Designers can use SOLIDWORKS 3D design to help others visualize their ideas, from 3D building engineering designs to prototype designs for new products, engaging demos, and more. The use of visual graphics helps the customer understand the features and specifications of the element. This also makes it easier to explain the mechanical component's utility to the client and prepare a proposal.

Conclusion

Additive manufacturing is a rapidly evolving field that offers unmatched customisation, cost-effectiveness, and design freedom. As technology advances and difficulties are resolved, and the whole potential of additive manufacturing is probably going to happen, spurring innovation in sectors. Companies that embrace and adapt to this transformative technology stand to gain a competitive edge in the rapidly changing global market.